Eukaryotes, Prokaryotes & the Scale of Cells

Answer sheet · 24 marks total

Section A — Eukaryotic or prokaryotic?

(a) Eukaryotic (1)   (b) Prokaryotic (1)   (c) Prokaryotic (1)   (d) Eukaryotic (1)

Section B — Bacterial cell structure

B1. Cytoplasm (1), cell membrane (1), cell wall (1).

B2. A small ring of DNA, separate from the main DNA loop (1).

Section C — Unit conversion

C1. 0.05×1000 (1)   = **50 µm** (1)

C2. $3000 \div 1000$ (1)   = **3 µm** (1)

C3. $2 \div 1000$ (1)   = **0.002 mm** (1)

Section D — Standard form

D1. 2 µm = 0.000002 m (1)   = **2×10^{-6} m** (1)

D2. 25 µm = 0.000025 m (1)   = **2.5×10^{-5} m** (1)

D3. 4×10^{-5} m = 0.00004 m (1)   = **40 µm** (1)

Section E — Order of magnitude

E1. $20 \div 2 = 10$ times bigger (1)   one order of magnitude (1)

E2. $(4 \times 10^{-5}) \div (2 \times 10^{-6}) = 20$ (1)   **20 times wider** (1)

Marking note: award method marks for correct working even if the final answer contains an arithmetic error. Accept any scientifically correct equivalent wording in Sections A and B.